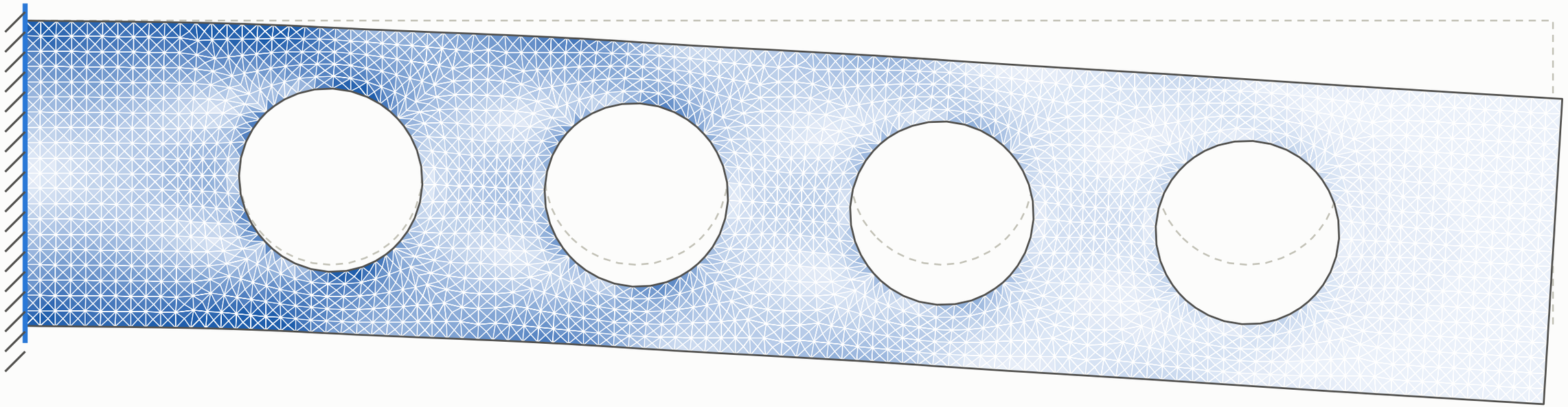
Minimum potential energy

Variational principles, the weak and the strong form, and linear elasticity in FEniCS



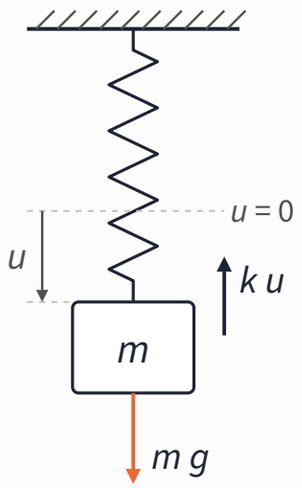
Variational principles in physics

Nature makes a functional stationary, often minimal. The physical law is the condition for it.

Physics regime		Variational principle	Strong form / physical law
Mechanics	Dynamics	**Stationary action** $\int \left(T - V\right)\, dt$ stationary	**Newton's law** $m\,\ddot{\boldsymbol{x}} = -\nabla V$
	Statics	**Minimum potential energy** THIS LECTURE $\Pi = \Psi - W$ minimal	**Balance of forces** $\sum \boldsymbol{F} = 0$
Optics		**Fermat's principle** travel time $\int \frac{n}{c}\, ds$ stationary	**Snell's law** $n_1 \sin\theta_1 = n_2 \sin\theta_2$
Electrostatics		**Dirichlet's principle** $\int \frac{1}{2}\,\epsilon\,\lvert\nabla\varphi\rvert^2 - \rho\,\varphi\, dx$ minimal	**Gauss's law** $-\nabla\cdot\left(\epsilon\,\nabla\varphi\right) = \rho$
Quantum mechanics		**Rayleigh–Ritz principle** $\langle\psi, H\psi\rangle \,/\, \langle\psi,\psi\rangle$ minimal	**Schrödinger equation** $H\psi = E\,\psi$

EXAMPLE

A mass on a spring



Balance of forces

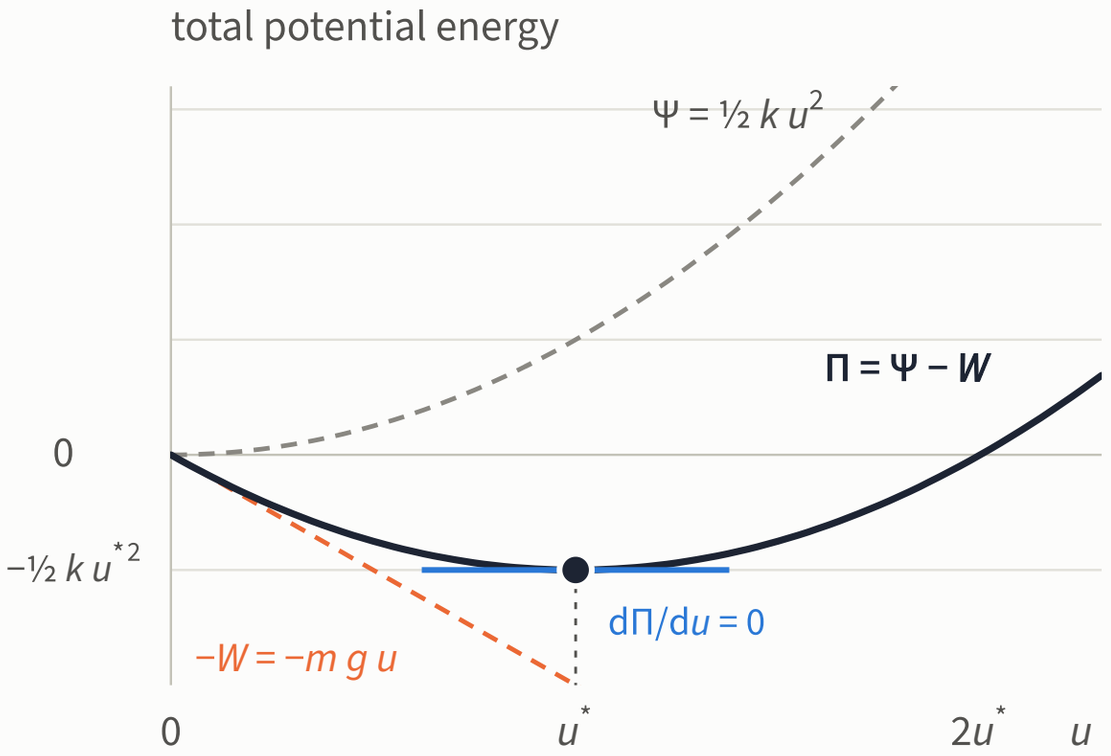
$$mg - ku = 0$$

hence $u^* = mg/k$

Minimum of the potential energy

$$\Pi(u) = \tfrac{1}{2}\,k\,u^2 - mg\,u$$

spring energy Ψ minus the work W of the weight



$d\Pi/du = ku - mg = -\sum F$: the minimum *is* the balance of forces.

The principle of minimum potential energy

A body in equilibrium takes the state u that **minimizes its total potential energy**

$$\Pi(u) = \Psi(u) - W(u)$$

among all **admissible** u: those that satisfy the essential boundary conditions.

Ψ: strain energy stored in the body, W: work of the applied loads

A symmetric, positive and linear

$$\Pi(u) = \tfrac{1}{2}\, a(u,u) - L(u)$$

with a and L of the variational formulation $a(u,v) = L(v)$.
Spring: $a(u,v) = k\,u\,v$, $L(v) = mg\,v$.

Why the minimizer solves $Au = f$

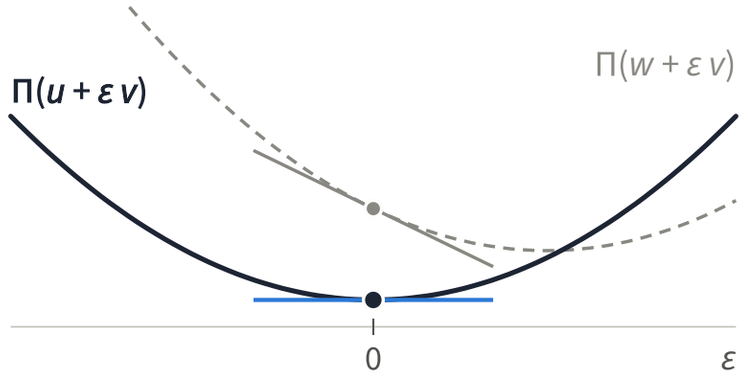
$$\Pi(v) = \tfrac{1}{2}\, a(v-u, v-u) - \tfrac{1}{2}\, a(u,u)$$

is smallest for $v = u$, with $\Pi(u) = -\tfrac{1}{2}\, a(u,u) = -\Psi(u)$

Directional derivative and Euler–Lagrange equation

Directional derivative

$$\delta\Pi(u;v) = \frac{d}{d\varepsilon}\,\Pi(u+\varepsilon v)\,\Big|_{\varepsilon=0} = a(u,v) - L(v)$$



$\delta\Pi(u;v) = 0$ for all v: the variational formulation, with the test function as the direction.

Euler–Lagrange equation

For $\Pi(u) = \int_\Omega \mathcal{L}(x, u, \nabla u)\, dx$:

$$\frac{\partial \mathcal{L}}{\partial u} - \nabla \cdot \frac{\partial \mathcal{L}}{\partial \nabla u} = 0$$

Heat conduction: $\mathcal{L} = \frac{1}{2}\,|\nabla u|^2 - f\,u$ gives $-\nabla \cdot \nabla u = f$.

One functional, the weak and the strong form

FUNCTIONAL

$$\Pi(u) = \int_\Omega \mathcal{L}(x, u, \nabla u)\, dx - \int_\Gamma t\, u\, ds$$

energy in Ω, work of the load t on the Neumann boundary Γ

directional derivative

Euler–Lagrange equation

VARIATIONAL FORMULATION

$$\int_\Omega \frac{\partial \mathcal{L}}{\partial u}\, v + \frac{\partial \mathcal{L}}{\partial \nabla u} \cdot \nabla v\, dx - \int_\Gamma t\, v\, ds = 0$$

for all v

integrate by parts

STRONG FORM

$$\frac{\partial \mathcal{L}}{\partial u} - \nabla \cdot \frac{\partial \mathcal{L}}{\partial \nabla u} = 0 \ \text{ in } \Omega$$

$$\frac{\partial \mathcal{L}}{\partial \nabla u} \cdot n = t \ \text{ on } \Gamma$$

Natural or Neumann

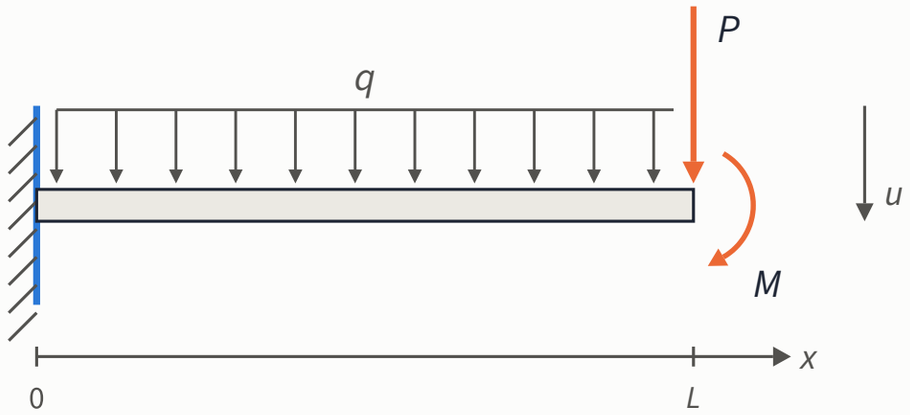
The boundary terms are part of Π: the natural conditions come out of the minimization.

Essential or Dirichlet

Not in Π: they restrict the admissible u (`DirichletBC`).

EXAMPLE

The biharmonic equation: a beam



Clamped at $x = 0$: $u(0) = u'(0) = 0$.
Load q, end force P, end moment M.

$$\Pi(u) = \int_0^L \tfrac{1}{2}\, EI\, (u'')^2\, dx - \int_0^L q\, u\, dx - P\, u(L) - M\, u'(L)$$

Directional derivative, the weak form with $s = 2$:

$$\delta\Pi(u; v) = \int_0^L EI\, u''\, v''\, dx - \int_0^L q\, v\, dx - P\, v(L) - M\, v'(L) = 0$$

EXAMPLE

The beam: strong form and boundary conditions

Integrate by parts twice ($v(0) = v'(0) = 0$):

$$\int_0^L \left(EI\,u'''' - q\right) v\,dx + \left(EI\,u''(L) - M\right) v'(L) - \left(EI\,u'''(L) + P\right) v(L) = 0$$

$$EI\,u'''' = q \;\text{ in } (0, L)$$

	$x = 0$	$x = L$
essential	$u = 0,\ u' = 0$	–
natural	–	$EI\,u'' = M,\ -EI\,u''' = P$

Lecture 1, exercise 2

The load vector there is $F_j = L(h_j)$ with $L(v) = P\,v(1) + M\,v'(1)$:

$$(F_3, F_4) = (P, M) = (1, 0)$$

The boundary terms we dropped there are in Π.

The functional and the Galerkin method

Insert $u_h = \sum_i U_i \, h_i$:

Galerkin (lecture 1): test with $v = h_j$

$$a(u_h, h_j) = L(h_j)$$

Ritz: minimize Π over V_h

$$\Pi(U) = \tfrac{1}{2} U^T K U - U^T F$$

Both give $K U = F$: the Galerkin method *is* the minimization of Π.

- u_h is the **best approximation** in the energy norm.
- $\Pi(U)$ is the function that CG minimizes (lecture 2).

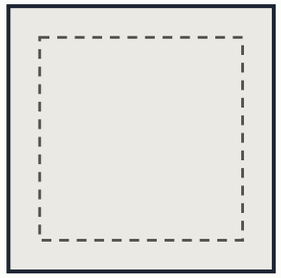
```
Pi  = (0.5*inner(grad(u), grad(u)) - f*u)*dx
dPi = derivative(Pi, u, v)     # delta Pi(u; v)
solve(dPi == 0, u, bc)
```

Displacement, strain and strain energy

The point x moves to $x + u(x)$; the **strain** is

$$\varepsilon(u) = \tfrac{1}{2}\left(\nabla u + \nabla u^T\right)$$

$\operatorname{tr}\varepsilon$: volume change, $\operatorname{dev}\varepsilon = \varepsilon - \tfrac{1}{3}(\operatorname{tr}\varepsilon)\, I$: change of shape



volume change
tr $\varepsilon \neq 0$, dev $\varepsilon = 0$
bulk modulus κ



change of shape
tr $\varepsilon = 0$, dev $\varepsilon \neq 0$
shear modulus G

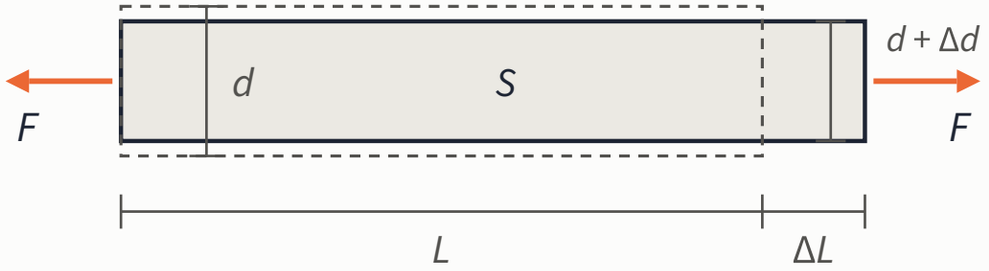
Strain energy density

$$\psi(\varepsilon) = \tfrac{1}{2}\,\kappa\,(\operatorname{tr}\varepsilon)^2 + G\,\operatorname{dev}\varepsilon : \operatorname{dev}\varepsilon$$

Stress

$$\sigma = \frac{\partial \psi}{\partial \varepsilon} = \kappa\,(\operatorname{tr}\varepsilon)\, I + 2G\,\operatorname{dev}\varepsilon$$

Measuring the constants



Tension test: the force F on the cross section S.

Young's modulus, Poisson's ratio

$$E = \frac{F/S}{\Delta L/L}, \qquad \nu = -\frac{\Delta d/d}{\Delta L/L}$$

Bulk and shear modulus (exercise 3):

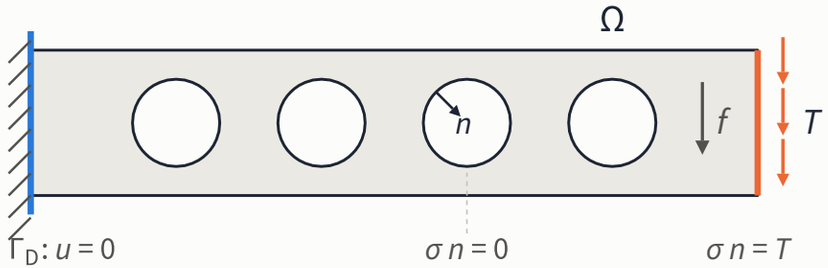
$$\kappa = \frac{E}{3\,(1-2\nu)}, \qquad G = \frac{E}{2\,(1+\nu)}$$

The examples use $E = 1000$ and $\nu = 0.3$.

The total potential energy of an elastic body

$$\Pi(u) = \int_\Omega \psi(\varepsilon(u))\, dx - \int_\Omega f \cdot u\, dx - \int_\Gamma T \cdot u\, ds$$

body force f, traction T on Γ, $u = 0$ on Γ_D



Directional derivative: the variational formulation of the dam (lecture 2)

$$\int_\Omega \sigma(u) : \varepsilon(v)\, dx = \int_\Omega f \cdot v\, dx + \int_\Gamma T \cdot v\, ds$$

Integrate by parts: the strong form

$$-\nabla \cdot \sigma(u) = f \text{ in } \Omega, \qquad \sigma n = T \text{ on } \Gamma$$

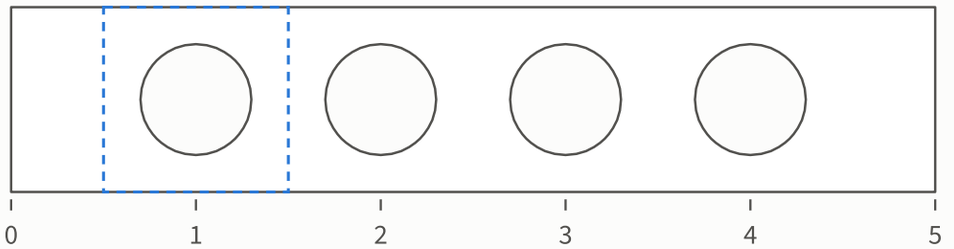
A beam with holes, without a mesh generator

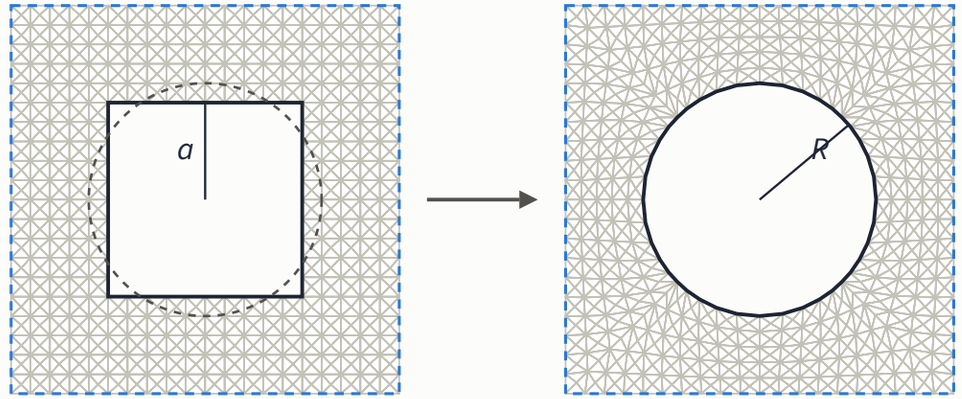
```python
from fenics import *
import numpy as np

length, height = 5.0, 1.0
R, a = 0.3, 0.25     # hole radius, half-width of the square cut out

box  = RectangleMesh(Point(0.0, 0.0), Point(length, height),
                     100, 20, 'crossed')
keep = MeshFunction('size_t', box, 2, 1)
for c in cells(box):     # cut out a square around each hole
    x, y = c.midpoint().x(), c.midpoint().y()
    if 0.5 < x < 4.5 and max(abs(x - round(x)), abs(y - 0.5)) < a:
        keep[c] = 0
mesh = SubMesh(box, keep, 1)

for X in mesh.coordinates():     # map the squares onto circles
    if 0.5 < X[0] < 4.5:
        centre = np.array([round(X[0]), 0.5])
        d = X - centre
        s, r = abs(d).max(), np.hypot(*d)
        t = (s - a)/(0.5 - a)    # 0 on the hole, 1 on the cell edge
        X[:] = centre + d*((1 - t)*R/r + t*0.5/s)
```





cut out: a square of half-width *a*

mapped: a circle of radius *R*

The square hole becomes a circle, the cell edges stay.
6400 triangles.

Shared setup: material, strain energy, boundaries

```
E, nu = 1000.0, 0.3
rho_g = 0.3
kappa = E/(3*(1 - 2*nu))    # bulk modulus
G     = E/(2*(1 + nu))      # shear modulus

V = VectorFunctionSpace(mesh, 'Lagrange', 1)

def epsilon(u):
    return sym(grad(u))

def psi(eps):               # strain energy density, plane strain
    return 0.5*kappa*tr(eps)**2 + G*(inner(eps, eps) - tr(eps)**2/3)

tol = 1e-3
def left(x, on_boundary):
    return on_boundary and near(x[0], 0.0, tol)

def right(x, on_boundary):
    return on_boundary and near(x[0], length, tol)

boundaries = MeshFunction('size_t', mesh, 1, 0)
AutoSubDomain(right).mark(boundaries, 1)       # ds(1): right end
ds = Measure('ds', domain=mesh, subdomain_data=boundaries)
```

Every example

1. write down `Pi`
2. `dPi = derivative(Pi, u, v)`
3. `solve(dPi == 0, u, bcs)`

Plane strain: $\varepsilon_{zz} = 0$, so `psi` takes the deviator in 3D.

Dirichlet – Dirichlet: a tension test

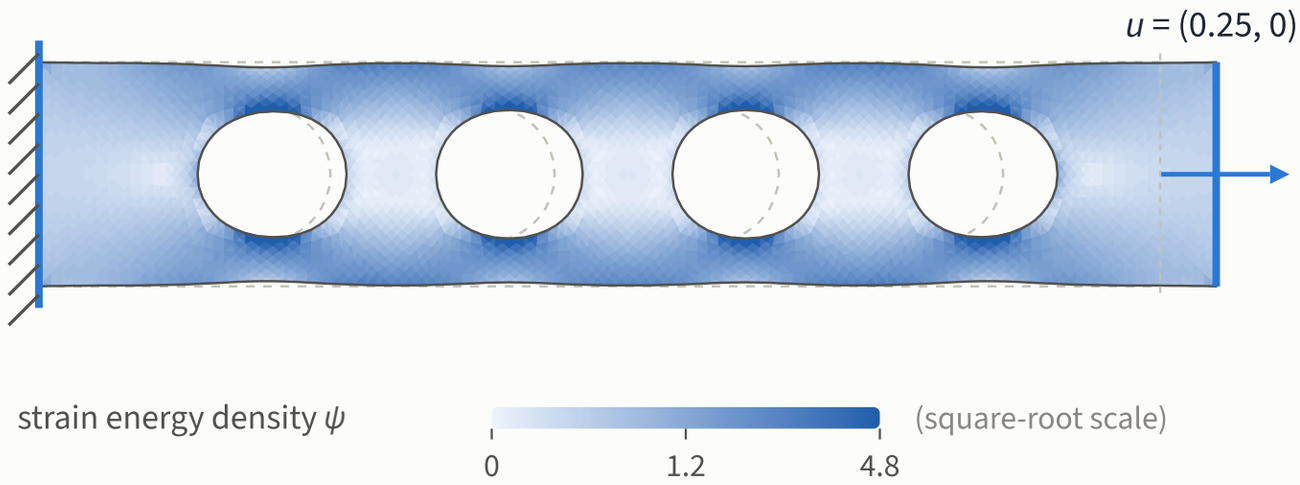
Left end fixed, right end pulled by 0.25; no loads: $\Pi(u) = \int_\Omega \psi(\varepsilon(u))\, dx$

```
bc0 = DirichletBC(V, Constant((0.0, 0.0)), left)
bc1 = DirichletBC(V, Constant((0.25, 0.0)), right)

u = Function(V)
v = TestFunction(V)

Pi  = psi(epsilon(u)) * dx
dPi = derivative(Pi, u, v)
solve(dPi == 0, u, [bc0, bc1])

print(assemble(Pi))     # stored strain energy
```



Stored energy 3.62, without the holes 6.94: the holes halve the stiffness.

Dirichlet – homogeneous Neumann: own weight

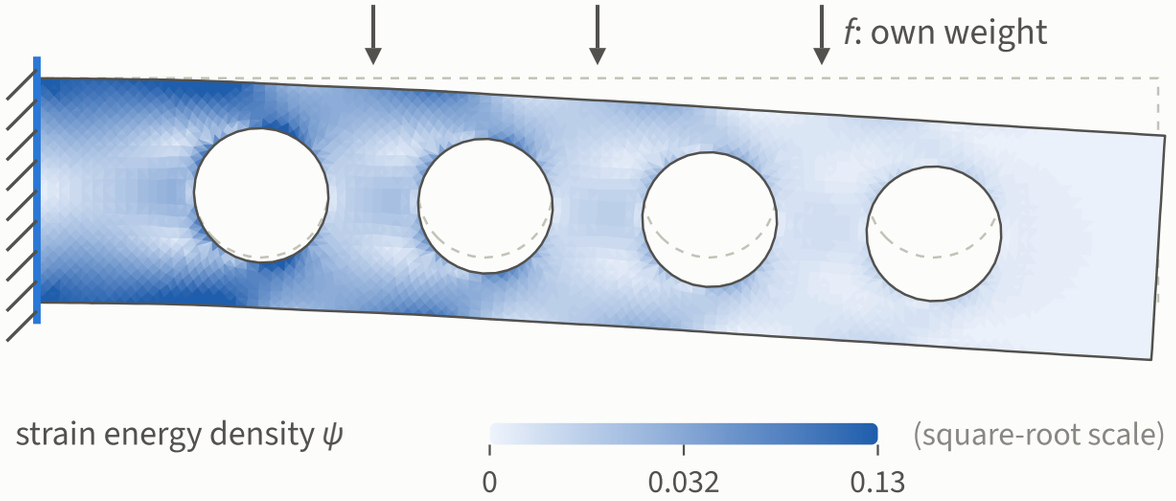
Left end clamped, body force $f = (0, -\rho g)$; no boundary term in Π.

```
bc = DirichletBC(V, Constant((0.0, 0.0)), left)
f  = Constant((0.0, -rho_g))

u = Function(V)
v = TestFunction(V)

Pi  = (psi(epsilon(u)) - dot(f, u)) * dx
dPi = derivative(Pi, u, v)
solve(dPi == 0, u, bc)

print(u(length, 0.5)[1])     # tip deflection
print(assemble(Pi), assemble(psi(epsilon(u))*dx))
```



Tip deflection -0.257, without the holes -0.265.

$\Pi(u) = -\Psi(u) = -0.064$, as for the spring.

- $\sigma n = 0$ on the holes and edges: no code.

Dirichlet – Neumann: an end load

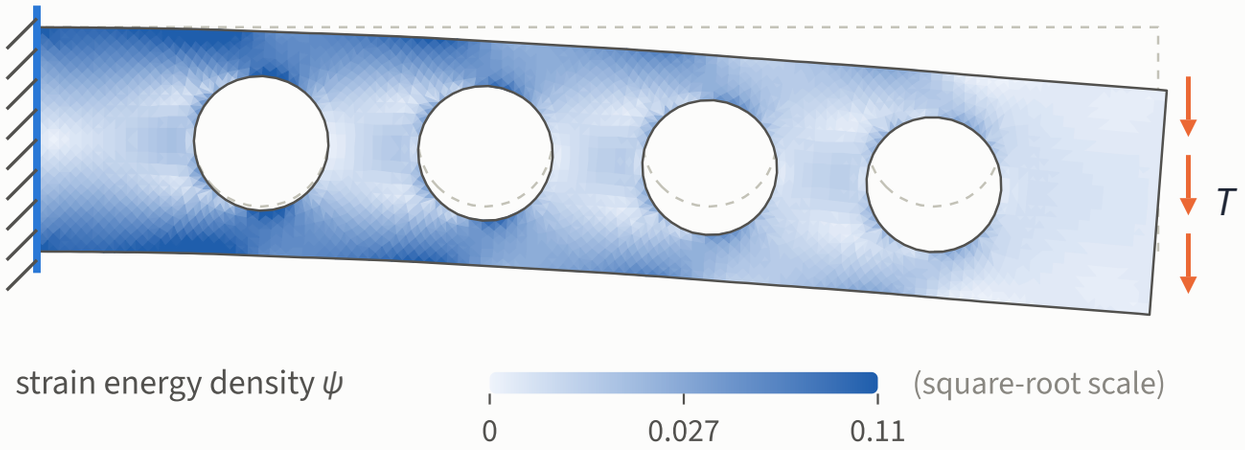
Left end clamped, traction $T = (0, -0.5)$ on the right end: the boundary term `ds(1)`.

```
T = Constant((0.0, -0.5))

u = Function(V)
v = TestFunction(V)

Pi  = psi(epsilon(u)) * dx - dot(T, u) * ds(1)
dPi = derivative(Pi, u, v)
solve(dPi == 0, u, bc)

print(u(length, 0.5)[1])     # tip deflection
print(assemble(Pi), assemble(psi(epsilon(u))*dx))
```



Tip deflection -0.282, without the holes -0.233: 21 % softer.
$\Pi(u) = -\Psi(u) = -0.071$.

Minimum potential energy in four steps

1. Write down $\Pi = \Psi - W$, with the work of all loads, also on the boundary.
2. $\delta\Pi(u; v) = 0$ for all v: the variational formulation (`derivative`).
3. Integrate by parts: the strong form and the **natural boundary conditions**.
4. Minimize over V_h: $K\,U = F$, the Galerkin method (`solve(dPi == 0, u, bcs)`).

EXERCISE 1

The functional of the heat conduction problem

Problem of lecture 1: $-\nabla \cdot \nabla u = f$ in Ω, u prescribed on $\partial\Omega \setminus \Gamma$, outward heat flux $q \cdot n = g$ with $q = -\nabla u$ on Γ. Consider

$$\Pi(u) = \int_\Omega \tfrac{1}{2} \left|\nabla u\right|^2 \, dx - \int_\Omega f\, u \, dx + \int_\Gamma g\, u \, ds.$$

1. Compute $\delta\Pi(u; v)$ and compare with the variational formulation of lecture 1.
2. Integrate by parts and derive the strong form and the natural boundary condition on Γ. Why does the boundary term of the functional carry a plus sign?
3. Solve problem 3 of lecture 1 ($u(0) = 0$, $f = 0$, $g = -1$ at $x = 1$, 20 elements) as a minimization problem with `derivative` and compare with the solution from `a == L`.

A simply supported beam

A beam of length L and stiffness EI under the uniform load q, supported at both ends: $u(0) = u(L) = 0$, the ends can rotate.

1. Which boundary conditions are essential, which natural? Write down $\Pi(u)$ and the admissible functions.
2. Derive the strong form and the natural boundary conditions from $\delta\Pi(u; v) = 0$ by integrating by parts twice. What is the bending moment $EI\,u''$ at the supports?
3. Solve $EI\,u'''' = q$ with these boundary conditions. Give the deflection at midspan and the slope at $x = 0$.

Material constants (theory)

1. Show that ψ equals $\frac{1}{2}\,\lambda\,(\operatorname{tr}\varepsilon)^2 + \mu\,\varepsilon : \varepsilon$ with $\lambda = \kappa - \frac{2}{3}G$, $\mu = G$, and compute $\sigma = \partial\psi/\partial\varepsilon$.
2. In the tension test $\sigma = \mathbf{diag}(\sigma_{xx}, 0, 0)$. Show that $\varepsilon = \mathbf{diag}(\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{yy})$ with

$$E = \frac{\sigma_{xx}}{\varepsilon_{xx}} = \frac{9\,\kappa G}{3\kappa + G}, \qquad \nu = -\frac{\varepsilon_{yy}}{\varepsilon_{xx}} = \frac{3\kappa - 2G}{2\,(3\kappa + G)}.$$

3. Invert the relations to obtain κ, G and λ from E and ν, and compute the values for $E = 1000$, $\nu = 0.3$. What happens for $\nu \to \frac{1}{2}$?